Knowledge Graph-Based AI Framework for Predicting Nutritional and Health Impacts of Food Ingredients

Project ID : 25-26J-094

Project Proposal Report

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BSc (Hons) Degree in Information Technology

Specialized in Data Science

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DECLARATION

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ABSTRACT

The increasing sophistication of contemporary food products requires intelligent, data-driven systems, able to analyze nutritional content, health remediation and risk, and potential allergens. Traditional approaches to labeling products often fail at detailed levels with respect to risks, and the complex interactions between nutritional categories. This paper presents an AI-based system that implements Knowledge Graphs (KGs), machine learning (ML), and graph neural networks (GNNs) to allow systematic predictions of health risks and allergic sensitivities related to food ingredients. By utilizing heterogeneous datasets from sources, such as USDA FoodData Central, PubMed, and FDA recalls, this system will create real-time Knowledge Graphs that model relationships between ingredients, health effects, and nutrients.

Graph Neural Networks aid in learning complex associations allowing for predictive modeling to assess risk for hypertension, obesity, and allergic responses. Central techniques—embeddings, relational reasoning, and modern ML algorithms—extract patterns and infer valuable relationships. Knowledge Graphs work with Large Language Models (LLMs) to summarize evidence from the literature based on the databases and supporting literature to maximize transparency and explainability.

Lastly, to encourage accessibility and user experience, the framework has an AI-enabled voice-supported interface that offers verbal nutritional analyses and insights to allergy information to provide information relevant to a user who is visually impaired. By synthesizing predictive analytics, knowledge representation, and interactive AI user interfaces, this framework produces actionable insights associated with health and allergen-related consequences of modern packaged food products by integrating complex nutritional data into actionable insights.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Networks
XAI	Explainable AI
RNN	Recurrent Neural Networks
ML	Machine Learning
ViT	Vision Transformer
YOLO	You Only Look Once
R-CNN	Region-based Convolutional Neural Networks
LIME	Local Interpretable Model-agnostic Explanations
GPU	Graphics Processing Unit
GNN	Graph Neural Networks
LLM	Large Language Model
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
ETL	Extract, Transform, Load
TTS	Text-to-Speech
NLP	Natural Language Processing
KG	Knowledge Graph
JSON	JavaScript Object Notation
API	Application Programming Interface
RDF	Resource Description Framework
SPARQL	SPARQL Protocol and RDF Query Language

GUI	Graphical User Interface
NER	Named Entity Recognition
CSV	Comma-Separated Values
LLaVA	Large Language and Vision Assistant
LayoutLMv3	Layout Language Model version 3
FAISS	Facebook AI Similarity Search

1 INTRODUCTION

1.1 Background Literature

Food and dietary supplement labels represent an important part of consumer informed choice and public health. The current nutrition labeling laws have loopholes such that food and dietary supplement labels are being made using vague nutrition claims, allergens which are not identified, misleading statements, or inferior food options that confuse or mislead consumers. Besides adding more complexity to the modern day food products, consumers are being more sensitive to health and sustainability as they attempt to make informed choices about their diets. It thus results in a compelling necessity of a labeling system that is more evident and smarter and that reports upon the nutritional worth and ecological effect of foodstuffs.

The systems used by food and dietary supplements labeling have been based on an archaic model of heavy manual validation and quiescent data and are failing to keep up with minimum standards to dynamic regulatory systems and consumer demand. The resultant effect is that customers are left with no idea of what the actual health impacts of their eating habits are, and the issue is compliance to the manufacturer as well as the trust and confidence of its clients.In the last few years, AI is developing with new computer vision, NLP, and knowledge graphs to open a new landscape in automating and improving the analytics of food labels. The frameworks and the ability to extract structured pieces of information from a label with visual richness and verify claims based on regulatory data, in addition to predicting health implications, have greatly improved.

The objective of the research is to establish a multidisciplinary, intelligent framework comprising AI-supported extraction of label content, verification of regulatory compliance, health impact assessment, and analysis of sociocultural trends. Integrating these elements into a coordinated ecosystem, it is anticipated that this framework will facilitate transparency, safety and personalization in food and dietary labeling, providing consumers and regulators trustworthy evidence-informed knowledge.

1.2 Literature Survey

Such enhanced review summarizes the major themes, methodological progress, and long-term issues surrounding the implementation of computational techniques, specifically, the application of knowledge graphs (KGs) and artificial intelligence (AI) to the field of personalized nutrition, food recommendation and ingredient substitution. Following the increase in the global health burden that is ascribable to dietary habits that encourage obesity, diabetes, and cardiovascular disease, the need to have intelligent systems that can help decompose the complex relationship between food constituents, nutrition profiles, and health outcomes on an individual basis is high. Traditionally, nutritional databases have been abundant in quantitative information but lacked connection and relational profundity and therefore were not useful to predictive analytics. Using more recent sources as a background, this review predicts the role of KGs as a supporting structure necessary to integrate AI to be able to make proactive and evidence-based interventions instead of descriptions. It investigates a dozen seminal studies to demonstrate the paradigm shift between the conventional data repositories to graph-based that are dynamically powered ecosystems that facilitate real-time forecasting of health-risks and tailored dietary advice.

The main issue of the use of AI in food science is the overwhelming volumes of data that have a heterogeneous origin, comprising both structured nutrition tables and other unstructured data, including clinical abstracts and recipes made by users. Knowledge graphs become a key enabling technology, providing the unified, semantically rich system of organizing the entities, including particular ingredients (e.g., high-fructose corn syrup), like recipes, as well as nutrients (e.g., glycemic load). These entities are interconnected with multifaceted relationships, and the predicates include, but are not limited to, the ones like exacerbates insulin resistance. Such an organizational design makes interoperability with various data sources and new knowledge-extraction applications, such as automated ingredient replacement to counteract allergies and the proposal of individualized nutrition regimens that take genetic allergies into account, possible [1] [7]. Specifically, food-specific KGs compile information on cornerstone repositories, like the USDA National Nutrient Database, cataloguing more than 8,000 food items with a detailed micronutrient composition; FoodOn, open-source ontology which standardises food taxonomy across cross-study comparison; and FooDB, a large database of chemical compounds, bioactive molecules, and metabolic pathways found in foods [2] [15]. Large-scale recipe corpora, e.g. Recipe1M+, include one million annotated recipes with high-resolution images to facilitate visual-nutritional associations, whilst crowdsourcing sites like OpenFoodFacts provide diversity in products labelling around the world although with variation in the quality of data [1] [8].

On this base of data, researchers have created domain-specific KGs to support more complex use cases, and in that way, they have shown the flexibility of graph-based representations in the context of marrying nutritional science and AI-based personalization. An example of such a solution is FoodKG, which combines both textual and visual data of Recipe1M+ with nutrient vectors of the USDA and ontological hierarchies of FoodOn. The resulting KG assists with the higher-order query-answering tasks, including the inference of cardiovascular risk based on the sodium-potassium ratio in a meal, and drives recommendation engines based on the prioritisation of balanced macronutrient meals [1]. Likewise, RecipeKG extracts a multifaceted graph on Allrecipes.com, including more than recipes as well as ingredient lists but also social metadata such as user ratings, and healthiness indices that are calculated based on authoritative benchmarks, such as the USDA Dietary Guidelines to Americans, as well as the nutrient-profiling model used by the UK Food Standards Agency to rank products on a scale, pitting healthy and unhealthy ingredients [7]. The Swiss Food Knowledge Graph (SwissFKG) is novel and contextualised locally by combining national dietary paradigms, including the Swiss Food Pyramid, detailed information on the use of specific ingredients in the region, some viable substitutes (e.g. almond milk instead of dairy in lactose intolerant profiles), daily reference intakes (DRIs), and allergen cross-reactivity maps. This helps to give culturally sensitive guidance that can help minimize non-compliance in the various groups of people [3]. These domain KGs are one of the better answers to the demand of living graphs, which update with evidence as it is introduced, an essential aspect to applications like the proposed framework, where regulatory changes, such as the FDA allergen changes, have to be disseminated dynamically to maintain prediction utility.

The complexity of food ecosystems as a self-organizing system with unique synergistic characteristics of ingredients, individual deviation, and changing health guidelines require more advanced AI methods that can be used to find successful substitution and recommendation paradigms. Early attempts into this space used methods of embedding vectors e.g. Word2Vec or the contextual word models proposed by spaCy which encode ingredient semantics into dense numerical vectors based on co-occurrence statistics across large collections of recipes, allowing intuitive substitutions e.g., clustering around quinoa with brown rice to give gluten-free alternatives, but lacking interpretability as the logic behind the relative proximity of vectors remains unclear [5][8]. These limitations have generated composite ranking schemes like the Diet-Improvement Ingredient Substitutability Heuristic (DIISH) which combines multiple similarity measures e.g. Word2Vec cosine distances, spaCy semantic matches, recipe-contextual Positive Pointwise Mutual Information (PPMI) on co-ingredient frequencies and pairwise compatibility scores into a composite ranking and can effectively leverage both latent patterns and explicit knowledge-graph semantics, with up to 20% increase of substitution relevance over single schemes in empirical experiments [9].

These methods can be further divided into more structure sensitive methods, including Knowledge -Graph Embeddings (KGEs) which map graph triples to continuous low - dimensional manifolds and can make predictions of a scale, like nutritionally equivalent substitutions, like embedding avocado oil next to olive oil in their monounsaturated fat space, without losing relational hierarchies. The recent state-of-the-art in this field is dominated by Graph Neural Networks (GNNs), including inductive variants like GraphSAGE to scale to node embeddings, and attention-based variants like Graph Attention Networks (GATs) to weighted relational aggregation; both are better at multi-hop reasoning on the basis of knowledge graphs, which is necessary to explain the indirect effects of trans fats on inflammation through the oxidative-stress-pathway. More applications in practice encompass GNN-based tools to substitute the Nutri-Score-enhanced tools, in which graph convolutions propagate nutritional scores among similar items to suggest healthier by design alternatives. On top of nutritional constraints, user preferences are also added to the complementary frameworks in contexts of health-aware recommendations using Graph Convolutional Networks (GCNs). By way of illustration, the Nutrition-Related Knowledge Graph develops a heterogeneous undirected graph of seven basic nutrients, including calories, total and saturated fats, sodium, protein, sugars, carbohydrates, and runs GCN layers to spread features, improving the score of dietary diversity and decreasing over-dependence on processed food in individualised plans. On the same note, the paradigm is re-energized in the Health-Aware Food Recommendation Model whereby TransD embeddings are augmented with attentional GCNs that are informed with knowledge and twin-task losses, one of them preference modelling and the other health constraints, are used to alternate outputs. Users who consume above threshold amounts of indicator nutrients, including sodium or saturated fats, are substituted under the model and UK FSA ratings applied, and the model is much more consistent with clinical recommendations in ablation studies.

At the same time, the stakes of nutritional interventions, whereby the sub-optimal advice might worsen such conditions as food allergies or chronic illnesses, have intensified the safety, transparency and adherence to guidelines imperatives [13]. Although large language models, such as ChatGPT, have revolutionized interfaces by their ability to understand natural-language extensions of meal databases, such as generating culturally variant alternatives, such as by reformulating a Mediterranean salad with South Asian spices without losing any anti-inflammatory properties, their propensity to hallucinations or fake facts that they source through uncurated web corpora are existentially dangerous, with error rates up to 30 per cent with ungrounded nutritional queries [12][13]. The new countermeasures have turned to the knowledge-graph-based retrieval with the paradigm of Graph-Retrieval Augmented Generation (Graph-RAG) on SwissFKG. This strategy pre-conditions LLM prompts with fact-checked graph traversal, which results in 25 40 35 improvements in factual accuracy and user confidence when querying the system with a query like: low-glycemic substitutes of rice in diabetic diets [3][13]. Transparency also splits methodologically; rule-based systems provide granular traceability, e.g. audit a recommendation by explicitly tracing, e.g. with edge paths, between a given ingredient, e.g. X, and an evidence-based outcome, e.g. Y [7][15] — whereas black-box embeddings do not have access to such nutritional details: post-hoc explanations based on SHAP values can be highly unspecific about an ingredient, e.g. SHAP 0.234.27. The rigor of sources determines trustworthiness: despite CMC making crowdsourced contributions of OpenFoodFacts democratised, interdisciplinary vetting is required, especially the involvement of nutritionists or clinicians, to eliminate biases or errors, which may spread dangerous alternatives [2][9]. In addition, the measures of healthiness should be more complex than mere aggregates; they should adopt multi-dimensional measurements, e.g., incorporating Glycaemic Index to assess the quality of carbohydrates in diabetic patients or modify the measurements to nutrient bioavailability in the elderly population [12][15].

Here, the trend converges to hybrid in the form of the neurosymbolic architectures that integrate the inductive properties of the deep learning in the form of embedding-based pattern discovery with the symbolic reasoning and rule-enforcement structures over knowledge graphs. These systems should deliver robust generalization, and explainability, which is a need of tools like yours that have to be legally compliant [1] [14].The suggested extensions can include sensory contexts, such as the molecular flavor graphs of palatability-preserving substitutes of FlavorDB, or disease ontologies of PubMed to support tailoring to precision-medicine, e.g. emphasising omega -3 in cardiovascular cohorts [8] [15].In order to speed up the development, it would be necessary to standardize the field by using curated sets of substitutions with gold health annotations, which would make cross-model benchmarking and reproducibility possible [5] [9].Overall, this study sheds light on how to incorporate dynamic knowledge graphs, graph-neural-network predictions, and voice-enabled explainability into your framework and also has provided an opportunity to pioneer work in underrepresented domains, such as real-time allergen prediction in changing global supply chains.

Feature / Capability	**[1]**	**[2]**	**[3]**	**[4]**	**[5]**	**Proposed Framework**
Knowledge Graphs for ingredient-health relationships	✓	✓	✓	✗	✓	✓
Graph Neural Networks for risk prediction	✗	✗	✗	✗	✓	✓
Multi-source data integration (USDA, FDA, PubMed)	✗	✓	✓	✓	✗	✓
Explainable AI outputs / evidence summarization	✗	✗	✗	✗	✗	✓
Voice-enabled AI chatbot interface	✗	✗	✗	✗	✗	✓
Regulatory compliance verification (FDA/USDA)	✗	✗	✗	✗	✗	✓
Real-time updates & dynamic KG	✗	✗	✗	✗	✗	✓

Table 1. 1: Comparison table

1.3 Research Gap

Food and dietary labels contain important information about essential nutritional information and potential allergens on food packages and restaurants. However, the current systems for assessing the direct and indirect health effects of foods are limited. Most evaluations rely on either manual interpretation or simple data extraction strategies that do not consider how multiple ingredients interact with one another over time or consider the effect of multiple ingredients aggregated into single food products. For example, a combination of high sodium, saturated fat added sugars can lead to cardiovascular or metabolic issues, but typical evaluations do not consider these multi-combinatorial factors.

Research has applied ideas from knowledge graphs and machine learning to model the potential health-media' impacts of ingredients, providing useful predictions related to health and health conditions such as hypertension, diabetes, or obesity. However, much of the work is still algorithmic, static, and relies on a pre-curated datasets of ingredient-combinations that are not dynamically updated with new findings in clinical settings, nutritional studies, or state and federal regulations. Consequently, predictions can be outdated, less than fully exhaustive, and not sufficiently linked to evidence.

Unfortunately, the predictive frameworks used do not provide robust interpretability. The predictive framework may predict potential forms of risk but failure to clearly explain the science behind it and to summarize available evidence in a way that researchers and agencies can use This deficit of transparency decreases confidence in the system and restricts its real-world applicability for decision-making by consumers, health professionals, or policy makers. Fourth, while multi-source data exists in the environment (e.g., PubMed, FDA recalls, USDA FoodData Central), the challenge remains to integrate the various datasets into a cohesive predictive model due to their levels of heterogeneity.

Studies typically fail to address the methodological issues in estimating the mixing of structured, semi-structured, and unstructured data with accuracy, explainability, or scalability. In conclusion, there is a materials gap in developing an empirically-tested and informatics-informed, dynamic and explainable predictive framework to model ingredient-to-health relationships, assist summarizing supporting scientific evidence, and provide translatable and evidence-based insights for regulatory and health impact evaluation purposes. Closing this gap is essential to elevate food label evaluations from simple nutritional reporting to actionable guidance specific to health consumer engagement.

1.4 Research Problem

Food and dietary supplement labels are meant to provide consumers with an understanding of how consumption of specific products can affect health; however, the labels do not provide any meaningful or actionable information about the potential health consequences of consuming a product. Labels often contain nutrition information and health claims, but expect consumers to evaluate the ingredients in a product separately or consider them individually when, in fact, they could have greater interactions between each other and their overall effects on human health. The lack of information can present multiple challenges to consumers, wellness professionals, and regulators assessing risk for diseases, such as cardiovascular disease, diabetes, obesity, and other adverse dietary-related health outcomes.

Current computational techniques have attempted to model ingredient-health relationships using machine learning and knowledge graph, but all current systems are largely static, fragmented, or un-scalable. Current knowledge graphs, for example, are built off unknown pre-curated datasets, do not reflect emerging scientific studies using different health diagnostic methods, do not reflect updated regulatory revisions or guidelines (e.g., dietary references, reports of the Dietary Guidelines Advisory Committee and/or other health-focused committees/agencies), or newly discovered ingredient-health associations, and predict health risk for consumers regardless of whether modifications to the prediction were substantiated by scientific evidence, contemporaneous, and/or interpretable evidence of health risk.

Further, frameworks do not provide mechanisms for transparent explanation, making it challenging to understand how a specific ingredient, ingredients, or their combinations are related to the predicted health outcome. Evidence does exist from various sources (e.g., PubMed, FDA recalls, and USDA FoodData Central), although data integration of this disparate evidence to develop a model that can be interpreted into a single entity is a challenge for future work. Without this integration, the production of scientifically valid health predictions able to come from food labels has limitations.

Thus, the research question is around developing a dynamic, comprehensive explanation of the framework used to assess health risk potential derived from food and dietary supplement labels. The framework should accurately model ingredient-health interactions, provide a summary of scientific evidence related to supporting health, and effectively communicate transparency and interpretability of the evidence to inform consumer decision-making and to guide regulatory review. This research question will address significant aspects to move from health predictions reliant on interpretation of a food label, toward evidence-based locus systems that empowers consumer decision-making to consider health outcomes.

2 RESEARCH OBJECTIVES

2.1 Main Objective

The main aim of this investigation is to come up with a broad, dynamic, and insightful structure that will foresee the future health consequences of food and dietary supplement products. The first stage consists of the forecast of health hazards of certain elements; this will require research into the make up of ingredients and simulation of complex interactions, including the synergy impacts of high sodium, saturated fat, and added sugars on cardiovascular and metabolic wellbeing. Graph Neural Networks and Knowledge Graphs are used to model the relationship between ingredients and health outcomes, therefore, the risks, both in the short term, and long term (e.g., hypertension, obesity, diabetes, and other diet-related illnesses). The manuscript, therefore, adds to the structure as it does not only report about nutrition and food products, but also on the provision of science-based evidence of the possible effects of consumption of commercially available items.

More importantly, the framework incorporates both heterogeneous scientific data, produces evidence-based, robust and up-to-date predictions, and integrates constantly new literature, newly formed associations between constituents and health effects, and regulatory advice. It relies on authoritative sources like the USDA FoodData Central, FDA recalls, the PubMed publications and clinical trials. To encourage explainability, the system encourages decipherable results, deciphers scientific data into a comprehensible format, and provides visualizations produced by AI, such as infographics that allow determining the specific risk factors concretely, making the methods practical to researchers and other interested parties without deep technical knowledge that would like to know how certain health risks were derived.

The framework interconnects scientific evaluation with regulatory evaluation and decision making by the consumers. In this regard, the regulators, through official health-risk evaluation guidelines, utilize scientific evidence to check food-label statements and enforce them, thus reducing the chances of misleading or unverified statements about foods and dietary supplements. Equally, the consumers are empowered by being able to clarify, have a summary of evidence, and being able to draw the visual representation of data, which allows them to make informed dietary decisions depending on the underlying data conveyed in the label statement.

Thus, the predictive model integration, ensuring that there is no source that has not been investigated, presents explainable data and approaches that can be regulated, which makes this framework a holistic and scientific-based approach of assessing health claims of food and dietary products. It deals with the inadequacies on nutritional risks assessment and labeling, including predictive ability, the level of justification, transparency of assessment, and a clear understanding of insights of dietary data

2.2 Sub Objectives

1. Ingredient-Health Relationship Modeling:

This sub-objective is intended to build Knowledge Graphs that are holistic and which are integrated to relate ingredients, nutrients and health outcomes. Graph Neural Network will be used to learn complex interactions and synergistic interactions between various ingredients and, as such, enable the modeling of both longitudinal and short-term health risks. The aim of this modeling is to form the basis of risk predictions involving allergic reactions, cardiovascular diseases, diabetes, and obesity so that predictions of risks are tied to reported scientific associations.

2. Multi-source data integration:

Multi-source data integration (Dolbeare, 2012) is an alternative method for data integration, especially when information is collected from multiple sources.<|human|>Multi-source Data Integration Multi-source data integration (Dolbeare, 2012) refers to another approach of data integration particularly when the data is gathered through several sources.This sub-objective will involve the combination of structured, semi-structured, and unstructured data that will consist of the collection of authoritative information, such as the USDA FoodData Central, FDA recall data, PubMed, and clinical research, and, therefore, lead to the creation of evidence-based predictions. The system is programmed to be updated automatically with new scientific discoveries and regulatory changes to reduce the shortcomings of stagnant datasets. The heterogeneous data is preprocessed and standardized in a manner that enables its reliable and smooth integration into predictive models that are used to analyze health-risks.

3. The predictive modeling and risk estimation are the next ones.

This phase is the assembly of organized, semi structured, and unstructured information by trusted sources as USDA Fooddata central, FDA recall, PubMed and academic literature to be used in making valid and evidence based predictions. The constant transformations in the regulatory situation and advances in the field of science make the system change regularly thus avoiding the limitations of the fixed datasets. After intensive preprocessing and standardization it is possible to reliably combine and use heterogeneous source information by predictive models that determine health risk.

4. Explainable Artificial Intelligence and Evidence Summarization.

The aim of this sub-objective is to use machine learning and techniques of Graph Neural Network to prevent possible health hazards using ingredient composition of food products. The framework assigns the probability of adverse events based on evidence based scoring metrics and supports the predictions using historical data of health events and clinical evidence. Having the scientific rigor as the basis of the system, it can effectively measure health risks and fill informational gaps related to the possible danger of health posed by certain products or by some combinations of ingredients.

5. AI Chatbot with Voice Interaction:

This sub-objective is focused on creating an interactive AI chatbot that is able to present health impact knowledge through both text and voice output. Users can ask queries about ingredient information, health risk estimates, and scientific evidence using a conversational platform. The voice-enabled functionality provides equitable access to content for users with visual impairment or users who do not want to read to desired content. The chatbot also provides chances for predictions and evidence summaries and visualizations on risk in a user-centric format and allows for on-demand exploration of label information and supports informed health and dietary decisions..

3 METHODOLOGY

This study's methods begin with the broader process of gathering and preparing diverse food and dietary data to ensure the system has the capabilities to effectively analyze nutritional information, ingredients, and related health outcomes. Initially, the focus will be gathering quality information from reputable sources, including USDA FoodData Central for nutritional composition (e.g., carbohydrates, fats, proteins, vitamins, minerals), FDA recall lists to help identify the potential risk of having a particular ingredient, and PubMed or clinical studies for understanding the ingredient-health relationships. Data will be collected from structured data sources (table, spreadsheets) and unstructured text (scientific research articles, regulatory documents). Afterwards, a data preprocessing step will be conducted to remove inconsistencies, create a standardized format across the data, and extract relevant relationships from the dataset. Unstructured text will be analyzed to identify the key entities, such as ingredients, nutrients, and health outcomes, so the data can be prepared for more in-depth modeling and analysis.

Once the data is prepared for analysis, the next step involves understanding how ingredients interact/lead to health outcomes. In this case, a Knowledge Graph will be constructed to illustrate the ingredients, nutrients, and health conditions as interconnected entities, capturing direct relationships and indirect relationships. Each ingredient and nutrient will be connected to the appropriate health outcome and relationships will be encoded numerically via embedding techniques to be used in modeling. This method enables the framework to identify complex relationships, such as the joint impact of high sodium and saturated fats on cardiovascular health, which might be missed using standard methodologies. The framework is capable of reasoning about combinations of food products or ingredients and predicting potential health risks with further systematic understanding of the data in a structured way.

With the newly structured data, the method begins predicting health risks. Graph-based computational models will be applied to identify relevant patterns in the knowledge graph in order to estimate a product's likelihood for adverse events such as diabetes, hypertension or obesity. The framework will generate risk score estimates for overwhelmed, combinations of products or ingredients, supplying a scientific basis for risk assessment. In order to facilitate the process for stakeholders, future iterations of the framework may produce an explanation of the risk findings in the form of summaries of relevant supporting evidence retrieved from academic research articles, FDA reports, and clinical data, or visualizations, such as infographics, or ingredient-risk axis heatmaps, to help the end user easily understand which food components contribute most to predicted risk.

The final phase of the methodology transitions from a conception of predicting risk to an understanding of risk-based engagement to accessibility and practical experience. The predicted risk and supporting evidence are being incorporated into an interactive AI (Artificial Intelligence) chatbot capable of text-based and speech-based communication. Users are able to ask about the risk of the ingredient, potential health effects, and regulatory aspects through the conversational interface. In response, the chatbot will provide explanations and appropriate visual summaries in real-time, ensuring transparency in produced outputs. This design also ensures accessibility by enabling voice capabilities for users with visual impairments in addition to creating a cognitive-friendly platform that can make sense of complex health-related data. It even cross-checks the predictions against relevant regulatory guidelines to help users notice inconsistencies between what might be a 'brand claim' on the label and the characteristics of the food product, while supporting the claim of advertising compliance with FDA (Food and Drug Administration) and USDA (United States Department of Agriculture) standards.

The approach involves continued validation and updates at all stages to ensure applicability and reliability. The risk-based predictions are validated against health-based historical datasets and clinical evidence, while usability studies help ensure outputs that facilitate interpretation. Knowledge Graphs and predictive models are updated continuously based on new research, ingredient-health aspect relationships, and changes in regulatory policies. Updates to the Knowledge Graphs and the prediction model support the continued actionability of the framework with regard to the translation of raw food label data into health-related guidance for the consumer.

3.1 System Architecture Diagram

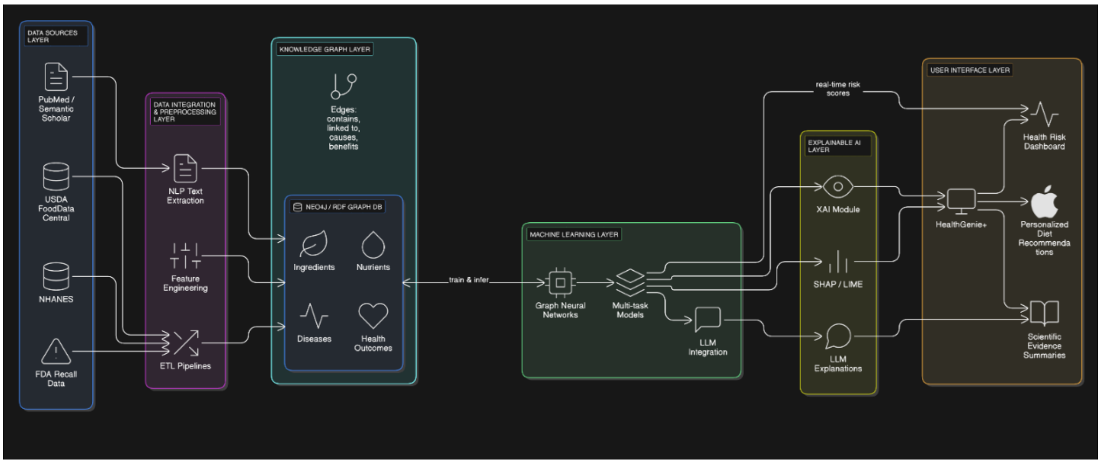

Figure 3. 1 : System architectural diagram

The proposed Knowledge Graph-Based AI Framework for Predicting Nutritional and Health Impacts of Food Ingredients is designed as a modular, scalable system that integrates the layers of data ingestion, knowledge representation, predictive analytics, explainable AI, and user interaction. Users shall interact with the system using an easy web or mobile interface where they can upload food labels (e.g., via image scan or text input) or query certain ingredients/products. In this way, user input is processed in real time, leveraging cloud-based services for computation with efficiency.

The core is the Data Ingestion Layer, which consolidates heterogeneous data from authoritative sources, including but not limited to nutritional profiles from the USDA FoodData Central, allergen and risk alerts from FDA recall databases, and evidence-based studies from PubMed/clinical repositories. For this layer, ETL pipelines are used, and integrated NLP tools, such as entity recognition via spaCy or BERT, can handle both structured data formats, like CSV/JSON, and unstructured data, like PDFs/articles.

The Knowledge Graph Construction Layer transforms the ingested data to a dynamic knowledge graph, either in Neo4j or in RDF triples, where nodes represent entities such as ingredients, nutrients, and health outcomes (e.g., hypertension or allergies) and edges encode relationships. An example is "high_sodium → increases_risk → cardiovascular_disease" with the weights of evidence. Embeddings (e.g., via GraphSAGE) are generated to capture multi-hop interactions, such as synergistic effects of sodium + saturated fats on obesity.

The Predictive Modeling Layer applies GNNs to learn patterns from input data and generate risk scores-e.g., probabilistic outputs about adverse health impacts for both the short and long run-using a library such as PyTorch Geometric. This layer outputs quantified risks-e.g., 75% likelihood of allergic response-validated against historical datasets.

The Explainable AI Layer integrates LLMs, such as GPT variants fine-tuned on domain data, for summarizing evidence; an example could be, "This risk is supported by 15 PubMed studies from 2020-2025." It also produces visualizations like heatmaps of ingredient-risk contributions or infographics via libraries like Matplotlib/Plotly.

Finally, the User Interaction Layer features a conversational AI chatbot, using Rasa or Dialogflow, to support text and voice, via Google Speech-to-Text/TTS, with which the user would query, for example: "What are the obesity risks in this snack?" The outputs are verbal summaries, accessible to visually impaired users, which can be exported as reports accessible to regulators.

The system is deployed on AWS/GCP for scalability; the relational/NoSQL database-e.g., PostgreSQL + Neo4j-is used for persistence. Security is guaranteed, complying with GDPR/HIPAA through encrypted data flows and role-based access.

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

The system should allow users-consumers, health professionals, and regulators-to upload/input food product labels either by typing, scanning the image (OCR-supported), or listing ingredients directly through web-mobile interfaces.

The system should support the ingestion and parsing of label data in common formats, including scanned images (JPEG/PNG), text files, or structured inputs like JSON from apps, by extracting key entities such as ingredients, nutritional values, and allergen declarations.

The system shall integrate multi-source data feeds from USDA FoodData Central, FDA recall databases, and PubMed APIs to automatically pull and update nutritional profiles, health associations, and regulatory alerts on a regular basis, such as daily. The system shall build and query dynamic Knowledge Graphs using Neo4j or any similar technology that helps in representing relationships among ingredients, nutrients, and health outcomes, including multi-hop inferences, for example, "sodium + saturated fat → hypertension risk".

The system will use GNNs to compute predictive risk scores for health impacts, for example, probability of obesity: 70%, allergy response: high/medium/low, based on ingredient combinations and user-specific factors provided, such as age and pre-existing conditions. Explainable outputs must be generated by the system. Examples include evidence summaries such as "Risk supported by 8 PubMed studies, 2022-2025" and visualizations, for instance heatmaps of risk contributions and infographics via integrated libraries such as Plotly.

The system shall provide an AI chatbot interface-text and voice-enabled-that will allow for natural language queries, such as "Assess diabetes risk for this cereal." Responses shall include verbal narration via TTS for accessibility. The system shall verify the predictions against regulatory standards, such as FDA labeling guidelines, highlighting non-compliant claims, such as "low sugar" when the thresholds are exceeded and providing compliance reports exportable in PDF/CSV format.

The system shall store user sessions, predictions, and KG updates in a secure database to offer tracking of previous queries and enable personalized recommendations, such as "Similar products with lower risk". The system shall support batch processing for multiple products, e.g., meal plans, and real-time updates to the KG in case of new data ingestion to maintain predictions up to date with the latest evidence.

4.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the Knowledge Graph-Based AI Framework, ensuring it is efficient, secure, and user-friendly while handling real-world demands in nutritional and health prediction. These requirements are categorized below for clarity.

Performance

- The system shall process a single food label input (e.g., image or text) and generate risk predictions within 5 seconds under normal load.
- The AI models (GNN and LLM) shall achieve prediction accuracy >90% on benchmark datasets (e.g., simulated clinical trials) and evidence summarization coherence >4.5/5 (human evaluation).
- Knowledge Graph queries (e.g., relationship traversal for multi-ingredient interactions) shall respond in <2 seconds for graphs up to 10,000 nodes.

Scalability

- The system shall handle concurrent processing of up to 1,000 user queries per hour, with horizontal scaling via cloud services (e.g., AWS Auto Scaling Groups).
- Data ingestion pipelines shall support integration with high-volume sources (e.g., daily PubMed updates of 500+ articles) without degradation, using distributed processing (e.g., Apache Kafka for ETL).

Reliability & Availability

- The system shall maintain 99.5% uptime for core services (e.g., prediction engine, chatbot), with automated failover and monitoring via tools like Prometheus.
- Databases (e.g., Neo4j for KG, PostgreSQL for user data) shall include automated backup and recovery mechanisms, ensuring data integrity with <1% loss in failure scenarios.
- Error handling shall gracefully manage invalid inputs (e.g., unreadable labels) by providing fallback explanations without system crashes.

Usability

- The interface (web/mobile/chatbot) shall follow WCAG 2.1 guidelines for accessibility, including voice navigation and high-contrast visuals for users with disabilities.
- User interactions shall require <3 clicks/steps to obtain a full risk assessment and evidence summary, with intuitive visualizations (e.g., interactive heatmaps) interpretable by non-experts.
- The system shall support multilingual queries (e.g., English, Spanish) via NLP, with response times adapted for voice output (e.g., <150 words per minute for TTS).

Security

- All data transmissions (e.g., label uploads, API calls) shall use HTTPS/TLS 1.3 encryption, with compliance to GDPR/HIPAA for handling sensitive health data.
- User authentication shall employ multi-factor authentication (MFA) and role-based access control (RBAC), restricting regulators to compliance reports and consumers to personal insights.
- The system shall log and audit all predictions for traceability, with anomaly detection to prevent injection attacks on KG queries or LLM prompts.

Maintainability & Extensibility

- The architecture shall be modular (e.g., microservices for data ingestion, prediction, and UI), allowing independent updates to components like GNN models without full redeployment.
- The framework shall support easy addition of new data sources (e.g., future WHO guidelines) or health outcomes (e.g., mental health impacts) via configurable KG schemas.
- Codebase shall adhere to standards (e.g., PEP 8 for Python) with >80% test coverage, enabling future enhancements like federated learning for privacy-preserving updates.

4.3 Personal Requirements

The interdisciplinary teamwork utilizing the knowledge of the nutritional science, health informatics, and advanced data science fields will play an essential role in creating the suggested Knowledge Graph-Based AI Framework to predict the nutritional and health effects of food ingredients. First of all, to provide guidance on the system design, it is necessary to have control by professionals who have extensive experience in the field of dietary nutrition, clinical health outcomes, and adherence to regulations, namely, FDA/USDA regulations, to illuminate on ingredient-health interactions, allergens dynamics, as well as evidence-based risk factors. The knowledge graph as well as the predictive model should represent the real-world dietary conditions that interact with each other to affect chronic conditions like diabetes or hypertension. Knowledge graph, graph neural network, natural language processing, and explainable AI technical mentorship is equally important to assure reliability and scalability of models. Especially, a strong oversight of supervisors, co-supervisors, and domain experts will be required to ensure that standards are high in preparing data, algorithm development, and validation. Combining the knowledge of nutrition, health, and technology, in turn, will form the basis of a successful and reliable AI in food safety and personal health advice.

It is vital to have profound knowledge about the intersection between the fields of nutrition, health informatics, and AI technologies. The effective studies presuppose the thorough knowledge of nutritional science along with the modern AI methods. The food products require deep expertise of metabolic patterns, health outcome measurement and effects on wellness and disease risk since the products are associated with myriads of threats requiring investigation on the ingredient. To this end, a data science expert has to master deep learning models, feature-extraction methods, and model-evaluation procedures to handle more complicated relational data and provide accurate predictions.The proposed system realizes health objectives through an integration of contemporary AI methods with field-specific knowledge. Accordingly, through domain connection, the system offers accurate, actionable insights that let consumers, health professionals, and regulators understand dietary risks more effectively, with better mitigation.

4.4 Expected Test Cases

The proposed Knowledge Graph-Based AI Framework for the prediction of nutritional and health impacts of food ingredients will be accuracy, reliability, and robustness tested during its development and deployment process. The major test cases are described here, which are adapted to the core components: data ingestion, KG construction, prediction, explainability, and interaction.

Table 4. 1: Test Case 1 - Data Ingestion and Preprocessing Test

Objective	Test Case	Expected Outcome
Data Ingestion and Preprocessing Test	Upload nutritional label data in different formats (JSON, scanned images via OCR) and varying completeness levels. Verify that the system preprocesses this data (entity extraction, standardization, noise removal) correctly.	The system should successfully ingest, preprocess, and prepare data for further analysis without any errors.

Table 4. 2 : Test Case 2 - Knowledge Graph Construction Test

Objective	Test Case	Expected Outcome
Knowledge Graph Construction Test	Input a dataset containing ingredient-health associations with manually annotated relationships. Compare the system's constructed KG edges with ground truth annotations.	The system should correctly build and link entities in the KG with high accuracy, closely matching ground truth relationships.

Table 4. 3:Test Case 3 - Predictive Risk Classification Test

Objective	Test Case	Expected Outcome
Predictive Risk Classification Test	Provide test data containing different ingredient combinations (e.g., high sodium + sugars for hypertension risk). Verify that the GNN model correctly identifies risk categories.	The system should classify health risks accurately according to their type with minimal misclassification.

Table 4. 4:Test Case 4 - System Scalability and Performance Test

Objective	Test Case	Expected Outcome
System Scalability and Performance Test	Upload a large batch of product labels (e.g., 1000+) simultaneously to evaluate system performance under load.	The system should process all inputs efficiently without performance degradation or crashes.

Table 4. 5:Test Case 5 - Explainable AI (XAI) Output Test

Objective	Test Case	Expected Outcome
Explainable AI (XAI) Output Test	Select a test label with ingredient risks. Verify that the system provides a heatmap or highlighted visualization showing why the model predicted that risk level.	The system should generate visual explanations (highlighted factors/heatmaps) clearly indicating the basis for risk prediction.

Table 4. 6:Test Case 6 - System Reliability Test

Objective	Test Case	Expected Outcome
System Reliability Test	Provide multiple random label inputs (some with high risks, some low). Check if the system consistently gives correct and stable outputs over repeated runs.	The system should produce consistent prediction/classification results across repeated trials without significant variations.

Table 4. 7:Test Case 7 - Usability and Storage Test

Objective	Test Case	Expected Outcome
Usability and Storage Test	Upload labels via the user interface and verify whether processed data and results (risk scores, evidence summaries, visualizations) are stored in the system database correctly.	All processed results should be saved in the database in an organized format and retrievable for later use.

5 COMMERCIALIZATION

The proposed Knowledge Graph-Based AI Framework for Predicting Nutritional and Health Impacts of Food Ingredients would add significant value to food manufacturers, regulatory agencies such as FDA/USDA, health research centers, and consumer wellness platforms. Its key benefit will be the ability to correctly, consistently, and quickly analyze nutritional risks and allergen impacts, something often not possible with manual reviews of labels or static databases. In particular, the method may prove important for experts when testing the validity of health-related claims on the label, when newly observed ingredient-health relationships come to their attention, such as in the case of new studies on ultra-processed foods, and when studying the spread of diet-related risks of metabolic disorders and allergies that develop into public health crises. The system can provide policy makers and researchers with evidence that can be measured to make evidence-based decisions that address the issue of label compliance, dietary guidelines and long-term population health interventions.

In the market, this technology is set as a technical analysis tool and not a crude caloric meter. The system can be integrated into the review pipeline of food industry laboratories, supplement developers and governmental regulatory agencies to improve the efficiency and accuracy of their processes or reduce the time between insight in their review processes. It is especially beneficial in the situation of sudden recalls or emergent health issues in which it is absolutely essential to determine the degree of seriousness and possible economic liability of harmful ingredient mixtures. It, therefore, is an effective decision-support instrument and provides a wide range of new business applications in compliance consulting, customized nutritional services and agricultural consultancy, thereby strengthening.

6 SOFTWARE SPECIFICATIONS

Software Component	Purpose / Functionality	Key Features / Notes
Python	Core programming language for data processing, ML, and AI implementation	Extensive libraries for ML, NLP, and data visualization (NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch)
TensorFlow / Keras	Building and training deep learning models (CNNs, GNNs)	Supports neural network modeling for nutrition and health impact prediction
PyTorch / PyTorch Geometric	Graph Neural Networks implementation	Handles relational learning for Knowledge Graph-based predictions
Neo4j / RDF Triple Store	Knowledge Graph construction and management	Supports SPARQL queries and integration with ML models for health risk prediction
OpenCV / LLaVA / LayoutLMv3	Intelligent label content extraction and vision-language modeling	OCR and document parsing for structured label data extraction
LangChain + FAISS	Regulatory claim verification using RAG framework	Embedding-based retrieval for FDA/USDA compliance verification
BERT / RoBERTa / Transformers	NLP for sentiment analysis, claim detection, and social listening	Fine-grained entity recognition, trend analysis, and text classification
FastAPI / Airflow	API development and workflow orchestration	Real-time data ingestion and automated processing pipelines
Tableau / Power BI / Dash	Visualization of trends, health risks, and compliance insights	Dashboard creation for interactive reporting of nutrition, health, and social insights

Federated Learning Framework (e.g., Flower)	Privacy-preserving personalized prediction	Enables distributed learning without sharing raw user data
Kafka / Microservices	Real-time data streaming and processing	Ensures scalable, modular, and real-time updates across system modules

Table 6. 1 : Software specifications

7 BUDGET

Category	**Cost (LKR)**
Software & Licenses	18,000
Hardware & Consumables	6,000
Contingency	6,000
Total	30,000

Table 7. 1 : Estimated Budget Table

The table provides a concise breakdown of the estimated budget for the project. It allocates funds to three main categories: Software & Licenses at LKR 18,000, which covers the cost of software tools and licenses required for data processing, analysis, and modeling; Hardware & Consumables at LKR 6,000, which includes any necessary physical equipment or materials; and Contingency at LKR 6,000, set aside to address unforeseen expenses during the project. The total estimated budget for these components amounts to LKR 30,000, ensuring that essential resources are accounted for while maintaining flexibility for unexpected costs.

8 WORK BREAKDOWN STRUCTURE

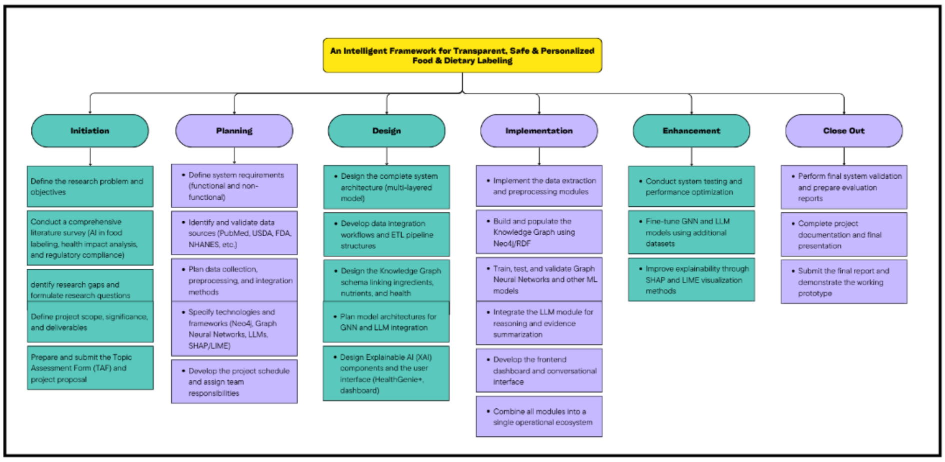

Figure 8. 1 : Work Breakdown Structure

The following Work Breakdown Structure (WBS) of the Knowledge Graph-Based AI Framework to Predict the Nutritional and Health Effects of Food Ingredients formally outlines the series of activities of the investigative research, as well as the system engineering. The project will start with the Initiation stage, where the general goals are defined and the key stakeholders (eg consumers, regulators and health professionals) are determined as well as pertinent nutritional and health risk groups (e.g., allergens, metabolic syndromes). The Planning stage is then implemented, and it includes a thorough requirement analysis, research agenda development, feasibility tests, and the clear acquisition of datasets in the official repositories like USDA FoodData Central and PubMed.

The Design stage describes the system architecture and gives algorithmic choices (i.e., graph neural networks to predictive modeling and Neo4j to build knowledge graphs). The stage also outlines preprocessing procedures and defines performance measures, e.g. accuracy levels at which to score risk. During the phase of Implementation, the research team gathers the data on nutritional labels and other forms of evidentiary data to train the model, and then undergoes strict validation, algorithm integration, and empirical testing with real-life product samples. The Enhancement stage is concerned with the efficacy of the models by including explainability capabilities, like large language model summaries, and automatic visualization of risk measures (e.g., heat maps). Lastly, the Close-Out phase officially ends the project by a thorough review exercise, which will deliver the documentation, spread of findings, and guidelines on future research including the possibility of integration with wearable health tracking devices.

9 GANTT CHART

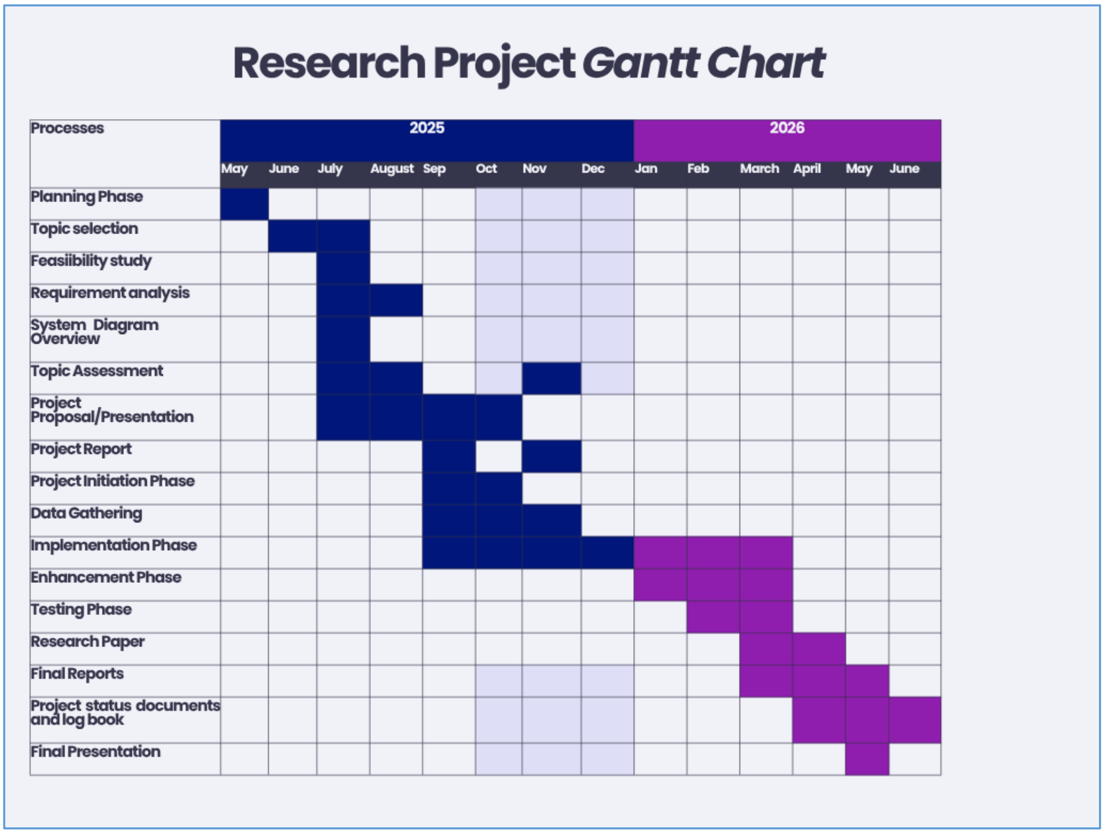

Figure 9. 1: Gantt Chart

The Gantt chart for the project is from May 2025 to June 2026 and covers all the stages of the project life cycle. It starts with the topic selection, then feasibility study, requirement analysis and system design in the early months. From September to December 2025, the project is on data collection and laying the foundation for the development stage. The implementation stage is from December 2025 to March 2026 where the proposed system is built, refined and optimized. Then enhancement stage to improve the system and address the shortcomings, after that testing from February to April 2026 to ensure accuracy and reliability. Finally the research paper is prepared, final report is completed and supporting outputs such as project website and research logbook are developed. The project ends with the final presentation and viva in May-June 2026. The Gantt chart is a high level overview of the tasks to ensure all activities are sequenced well and the project will run smoothly until completion.

10 CONCLUSION

This study proposes a novel and unified model of clear, safe and customized food and dietary labeling, which can solve the fundamental problems of the existing labeling traditions including false advertisements, hidden allergens and vague health outcomes. With the intelligent label content extraction, regulatory compliance validation, AI-based nutrition and health impact prediction, and social listening to analyze trends, the system presents a multidimensional solution, which will increase the degree of accuracy, transparency, and consumer trust.

The project illustrates how state-of-the-art methods, such as knowledge graphs, graph neural networks, large language models, and explainable AI can be integrated to create actionable insights using a wide range of sources and, as a result, close the divide between regulatory requirements, scientific research, and consumer demands. Individualised and evidence based recommendations and interactive voice interfaces will be incorporated to make their application accessible to all users, including those with defective eyesight and to aid in informed food choice among the entire population.

Moreover, this framework aids both manufacturers and regulators by offering automated compliance checks, trend analysis and sustainability check, which may inform product formulation, marketing, and policy. The scalable and modular nature of the system enables ongoing updates with the new scientific, regulatory, and social information, which makes it flexible to changing dieting trends and consumer demands.

To sum up, besides bringing technological innovation to the food labeling industry, the project enhances the health of the community, empowerment of consumers and sustainability. It provides a basis of future research and development in intelligent, explainable, and user-friendly food and dietary labeling systems, indicating the possibility of changes in the way consumers engage food products as well as help stakeholders make evidence-based decisions.

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